

Multi-Class Stroke Lesion Segmentation with Multi-Rater Consensus using a U-Net Framework

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1 Methods

All input volumes were first clipped to a fixed intensity range of $[-100, 300]$ HU to reduce outlier effects. Each 3D scan of size $\mathbb{R}^{512 \times 512 \times D}$ was then reshaped to match the network input format, resulting in a tensor of shape $\mathbb{R}^{1 \times 1 \times D \times 512 \times 512}$ (Batch $\times$ Channel $\times$ Depth $\times$ Height $\times$ Width). This ensures compatibility with the 3D U-Net architecture while preserving the original spatial resolution in the axial plane.

We employed a standard 3D U-Net architecture with a base channel size of 16. The network takes an input of shape $\mathbb{R}^{1 \times 1 \times D \times 512 \times 512}$ and outputs a segmentation map of shape $\mathbb{R}^{1 \times 6 \times D \times 512 \times 512}$, where each channel corresponds to one class: channel 0 represents background, and channels 1–5 correspond to the five different types of intracranial hemorrhage subtypes. The pipeline of the proposed system is shown in Figure 1.

To train the network for multi-class, multi-rater segmentation, we combined a masked Dice loss with multi-class cross-entropy. The Dice loss was computed per foreground class using one-hot encoded target and excluded classes absent in the current target volume, preventing instability from empty cases. The cross-entropy term was calculated by integer class indices along the channel dimension. The final loss was a weighted combination of Dice and cross-entropy. This approach allows the network to leverage inter-rater variability while accurately segmenting the five hemorrhage subtypes and background, and the approach is defined as:

$$\mathcal{L}_{\text{total}} = 0.7 \times \mathcal{L}_{\text{Dice}} + 0.3 \times \mathcal{L}_{\text{CE}}, \quad (1)$$

$$\text{Dice}_c = \frac{2 \sum_{b,d,h,w} p \times y_{\text{hot}}}{\sum_{b,d,h,w} (p + y_{\text{hot}}) + \epsilon}, \quad (2)$$

$$\text{mask}_c = \begin{cases} 1 & \text{if } \sum_{b,d,h,w} y_{\text{hot}} > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{\sum_{c=1}^{C-1} \text{Dice}_c}{\sum_{c=1}^{C-1} \text{mask}_c}, \quad (4)$$

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B \cdot D \cdot H \cdot W} \sum_{b,d,h,w} \sum_{c=0}^5 \mathbf{1}\{y_{\text{int}} = c\} \log p, \quad (5)$$

$$(6)$$

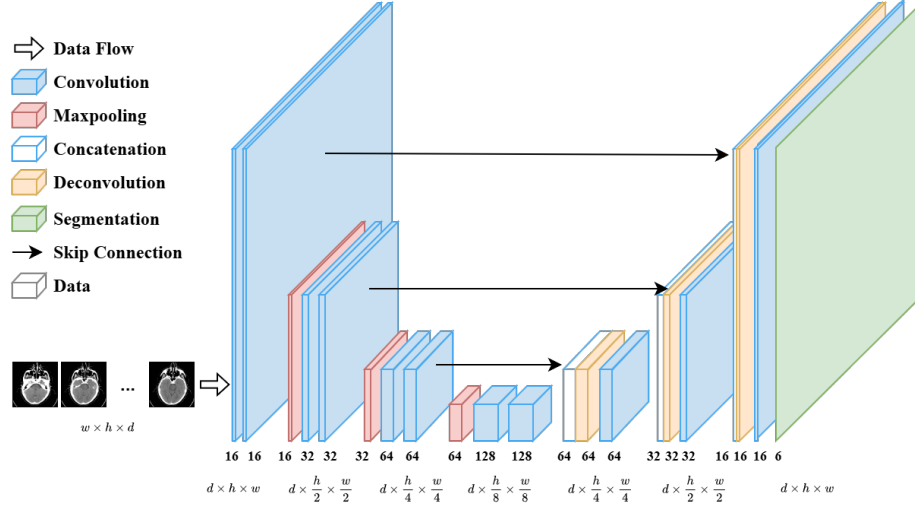


Fig. 1: Overview of the proposed MCMR-UNet. The proposed method is built based on U-Net [1] and generates 6 classes of output (0 for background, and 1-5 for 5 different hemorrhage subtypes).

where b, c, d, h, w are batch size, channels, depth, height, and width, p stands for $p_{b,c,d,h,w} = \text{softmax}(\text{logits})_{b,c,d,h,w}$ which represents the predicted probability, y_{hot} is the one-hot encoded target, $\epsilon = 10^{-6}$ prevents division by zero, $C = 6$ is the total number of classes, and $\mathbf{1}\{\cdot\}$ is the indicator function.

Model training was performed on a workstation equipped with an Intel Core i7-13700 CPU, an NVIDIA RTX 4090 GPU, and 32 GB of RAM, running Ubuntu 20.04, Python 3.9.18, CUDA 11.8, and cuDNN 8.7.0. The 3D U-Net models were implemented in PyTorch, with GPU acceleration enabled via CUDA 11.8. Training was conducted using the Adam optimizer with a learning rate of 1×10^{-4} . The network reached convergence after approximately 44 epochs.

References

1. Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: International Conference on Medical image computing and computer-assisted intervention. pp. 234–241. Springer (2015)